

Lecture 4: Inverse and Logarithmic Functions

An inverse function reverses a process. Addition and subtraction reverse each other; multiplication by a nonzero number is reversed by division. The exponential function b^x also has an inverse, but the operation that reverses “raise b to a power” needs a new name: the **logarithm**. This lecture develops inverse functions first, then uses that idea to make logarithms useful for solving exponential equations and interpreting real-world scales.

1. Inverse Functions and Their Graphs

A function f has an **inverse function** f^{-1} exactly when every output of f comes from one input. Such a function is called **one-to-one**. The inverse reverses input and output:

$$f^{-1}(f(x)) = x, \quad f(f^{-1}(y)) = y.$$

Horizontal line test. A function is one-to-one if and only if every horizontal line meets its graph at most once. A repeated y -value would mean two different inputs produce the same output, so an inverse could not decide which input to return.

Finding an inverse algebraically. To find f^{-1} :

1. Write $y = f(x)$.
2. Solve the equation for x in terms of y .
3. Swap the variable names x and y .
4. Write the result as $f^{-1}(x)$, and state its domain if it is restricted.

Example. Find the inverse of $f(x) = 5 - 2x$.

Write $y = 5 - 2x$, then solve for x :

$$2x = 5 - y \quad \Rightarrow \quad x = \frac{5 - y}{2}.$$

Now swap the variable names. Thus

$$\boxed{f^{-1}(x) = \frac{5 - x}{2}}.$$

Check both compositions:

$$f^{-1}(f(x)) = \frac{5 - (5 - 2x)}{2} = x, \quad f(f^{-1}(x)) = 5 - 2\left(\frac{5 - x}{2}\right) = x.$$

Graphs of inverses. If (a, b) lies on the graph of f , then (b, a) lies on the graph of f^{-1} . Swapping coordinates reflects a point across the line $y = x$, so the two graphs are reflections across $y = x$. Their domains and ranges exchange:

$$\text{dom}(f^{-1}) = \text{range}(f), \quad \text{range}(f^{-1}) = \text{dom}(f).$$

Restricted domains. The function $f(x) = x^2$ is not one-to-one on all real numbers because $f(2) = f(-2) = 4$. Restricting its domain to $x \geq 0$ makes it one-to-one. Then its inverse is

$$f^{-1}(x) = \sqrt{x}, \quad x \geq 0.$$

The restriction is essential: $\sqrt{x}$ returns the nonnegative square root only.

Exercise 1 (one-to-one). Use the horizontal line test or an algebraic reason to decide whether each function is one-to-one on the stated domain.

- $f(x) = 4x - 1$ on $\mathbb{R}$

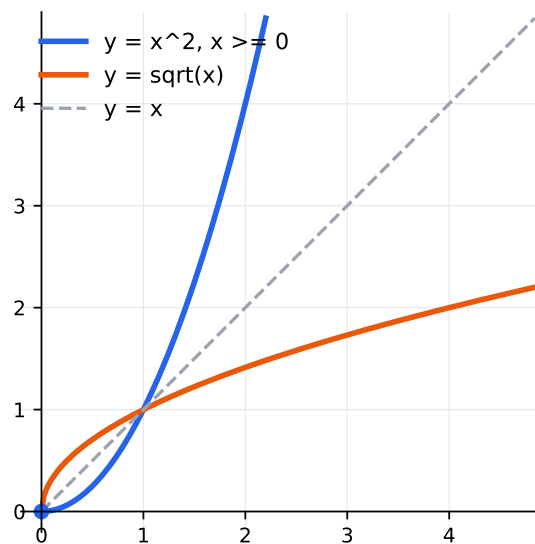


Figure 1: The graph of y equals x squared restricted to x at least 0 and its inverse y equals square root of x , reflected across y equals x

- $g(x) = x^2 - 3$ on $\mathbb{R}$
- $h(x) = x^2 - 3$ on $[0, \infty)$

Exercise 2 (inverse). Find the inverse of $f(x) = \frac{x+4}{3}$, then verify both compositions.

Exercise 3 (restricted inverse). Restrict $f(x) = (x-1)^2$ to $x \geq 1$. Find $f^{-1}(x)$ and state the domain and range of both f and f^{-1} .

2. Logarithms: Inverses of Exponentials

Let $b > 0$ and $b \neq 1$. The **logarithm of x to base b** is the exponent to which b must be raised to obtain x :

$$\boxed{\log_b x = y \quad \text{if and only if} \quad b^y = x.}$$

This equivalence is a translation rule, not a new calculation rule. For example,

$$\log_2 8 = 3 \quad \Leftrightarrow \quad 2^3 = 8,$$

and

$$\log_{10} \left(\frac{1}{100} \right) = -2 \quad \Leftrightarrow \quad 10^{-2} = \frac{1}{100}.$$

Because $b^x > 0$ for every real x , a logarithm accepts only positive inputs:

- Domain of $y = \log_b x$: $(0, \infty)$.
- Range of $y = \log_b x$: $(-\infty, \infty)$.
- x -intercept: $(1, 0)$ because $\log_b 1 = 0$.
- Vertical asymptote: $x = 0$.
- If $b > 1$, $\log_b x$ is increasing; if $0 < b < 1$, it is decreasing.

The notation $\log x$ means $\log_{10} x$ (the **common logarithm**), and $\ln x$ means $\log_e x$ (the **natural logarithm**). Since $\ln$ is the inverse of e^x ,

$$\ln(e^x) = x, \quad e^{\ln x} = x \quad (x > 0).$$

The functions $y = 2^x$ and $y = \log_2 x$ are inverses, so their graphs reflect across $y = x$. Notice how the horizontal asymptote $y = 0$ of the exponential becomes the vertical asymptote $x = 0$ of the logarithm.

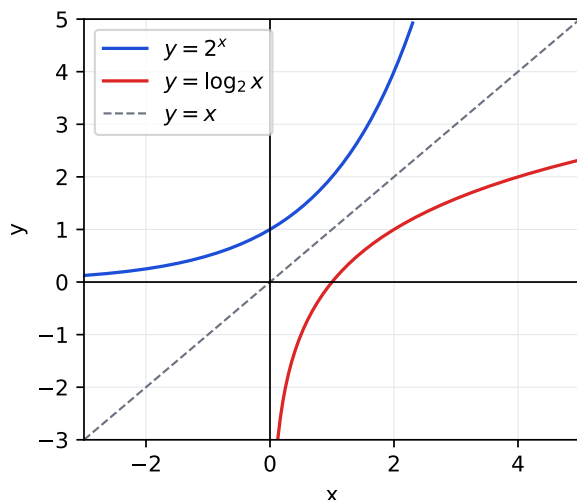


Figure 2: The graphs of y equals 2 to the x and y equals log base 2 of x reflected across y equals x

Example. Evaluate without a calculator:

$$\log_3 81 = 4, \quad \log_5 \sqrt{5} = \frac{1}{2}, \quad \log_2 \left(\frac{1}{8} \right) = -3.$$

Each answer is an exponent: $3^4 = 81$, $5^{1/2} = \sqrt{5}$, and $2^{-3} = 1/8$.

Exercise 4 (translation). Rewrite each statement in the other form.

- $4^3 = 64$
- $10^{-4} = 0.0001$
- $\log_7 49 = 2$
- $\ln(e^{-3}) = -3$

Exercise 5 (exact values). Evaluate without a calculator.

- $\log_2 32$
- $\log_9 3$
- $\log_{10}(0.01)$
- $\ln 1$

Exercise 6 (features). State the domain, range, x -intercept, vertical asymptote, and monotonicity of $y = \log_{1/3} x$, then plot the graph, labeling the asymptote.

3. Logarithm Rules and Equations

The logarithm rules come from the exponent rules. For positive M, N and real r ,

$$\log_b(MN) = \log_b M + \log_b N,$$

$$\log_b \left(\frac{M}{N} \right) = \log_b M - \log_b N,$$

$$\log_b(M^r) = r \log_b M.$$

For example, if $u = \log_b M$ and $v = \log_b N$, then $M = b^u$ and $N = b^v$. Thus $MN = b^u b^v = b^{u+v}$, so $\log_b(MN) = u + v$.

Example. Expand and simplify:

$$\log_2 \left(\frac{8x^3}{y} \right) = \log_2 8 + \log_2 x^3 - \log_2 y = 3 + 3\log_2 x - \log_2 y,$$

where $x > 0$ and $y > 0$.

Solving exponential equations. If both sides can be written with the same base, use equal exponents. Otherwise, take a logarithm of both sides. For $a > 0$, $a \neq 1$, and $C > 0$,

$$a^x = C \implies \ln(a^x) = \ln C \implies x \ln a = \ln C \implies \boxed{x = \frac{\ln C}{\ln a}}.$$

The last quotient is the **change-of-base formula**:

$$\boxed{\log_b C = \frac{\ln C}{\ln b} = \frac{\log C}{\log b}}.$$

Exercise 7 (log rules). Expand each expression as far as possible. Assume all variables are positive.

- $\log_5 \left(\frac{25x}{z^2} \right)$
- $\ln \left(\frac{\sqrt{x}}{3y} \right)$
- $\log_2(16a^4b)$

Exercise 8 (exact equations). Solve exactly.

- $3^{2x-1} = 27$
- $\log_2(x-1) = 4$
- $\ln(x+2) = 0$

Combining logarithms and checking for extraneous solutions. When an equation has more than one logarithm, use the product or quotient rule to combine them into a single logarithm, then convert to exponential form. Because a logarithm is defined only for a positive argument, **always check every candidate solution in the original equation** — combining logarithms can introduce a root that does not actually belong to the domain.

Example. Solve $\log_2(x+1) + \log_2(x-1) = 3$.

The original equation requires $x+1 > 0$ and $x-1 > 0$, i.e. $x > 1$. Combine the left side with the product rule, then convert to exponential form:

$$\log_2 [(x+1)(x-1)] = 3 \implies \log_2(x^2 - 1) = 3 \implies x^2 - 1 = 2^3 = 8 \implies x^2 = 9 \implies x = \pm 3.$$

Since $x = -3$ fails $x > 1$, it is **extraneous** and must be rejected. Only

$$\boxed{x = 3}$$

solves the original equation. (Check: $\log_2 4 + \log_2 2 = 2 + 1 = 3$.)

Exercise 9 (extraneous solutions). Solve each equation, then check every candidate solution against the domain of the original logarithms and discard any extraneous root.

- $\log_3(x+2) + \log_3 x = 1$
- $\log(x) + \log(x-3) = 1$

Exercise 10 (calculator equations). Use a calculator and round each answer to three decimal places.

- $2^x = 7$
- $5e^{0.4t} = 60$
- $\log_3(x+1) = 2.7$

4. Transformations of Logarithmic Functions

The same transformation rules used for exponentials apply to logarithms. For $y = A \log_b(x - h) + k$, the parent graph $y = \log_b x$ is shifted right h , shifted up k , vertically scaled by A , and reflected across the x -axis when $A < 0$.

The domain and vertical asymptote move together:

$$x - h > 0 \implies x > h, \quad \text{vertical asymptote } x = h.$$

The vertical shift k does **not** move the vertical asymptote.

Example. For $y = \log_2(x - 2) + 1$, start with $y = \log_2 x$, shift right 2, then up 1. The vertical asymptote shifts from $x = 0$ to $x = 2$, the domain is $(2, \infty)$, and the parent point $(1, 0)$ moves to $(3, 1)$.

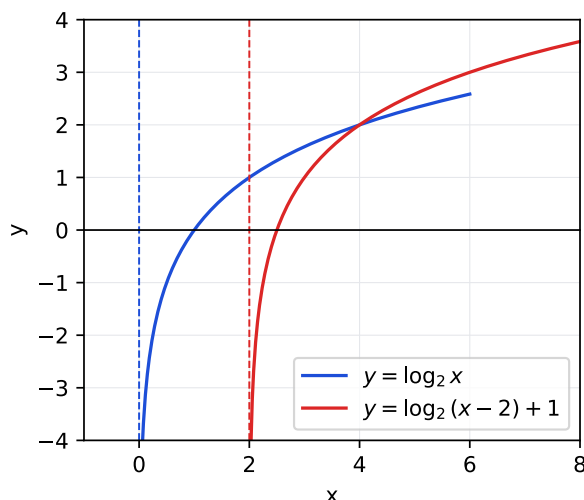


Figure 3: The parent logarithm y equals $\log$ base 2 of x and its right-2 up-1 shift, with vertical asymptotes x equals 0 and x equals 2

Reflected logarithms. Replacing x by $-x$ reflects the graph across the y -axis: $y = \log_b(-x)$ has domain $x < 0$, with the same vertical asymptote $x = 0$. More generally, $y = \log_b(h - x)$ has domain $x < h$ and vertical asymptote $x = h$ — the mirror image of $y = \log_b(x - h)$.

Exercise 11 (features). For $y = -2 \log_3(x + 1) + 4$, describe the transformations from $y = \log_3 x$, then state the domain, vertical asymptote, and the point corresponding to $(1, 0)$.

Exercise 12 (sketch). Sketch $y = \ln(x - 3)$, labeling its vertical asymptote and two points obtained from $y = \ln x$.

Exercise 13 (challenge). Explain why $y = \log_2(x^2)$ is not the same function as $y = 2 \log_2 x$. State the domain of each function and identify where their formulas agree.

5. Logarithms in Real Life

Logarithms make an enormous range manageable by recording an exponent instead of the raw quantity. An increase of 1 on a base-10 log scale means the original quantity is multiplied by 10.

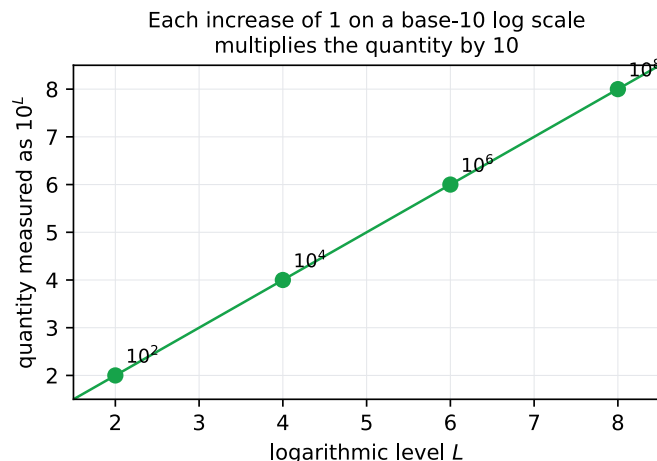


Figure 4: A base-10 logarithmic scale showing that each increase of 1 represents multiplication by 10

pH. The pH of a solution is defined by

$$\text{pH} = -\log_{10}[\text{H}^+],$$

where $[\text{H}^+]$ is the hydrogen-ion concentration in moles per liter. A solution with $[\text{H}^+] = 10^{-3}$ has pH 3, while one with $[\text{H}^+] = 10^{-5}$ has pH 5. The latter has 100 times *less* hydrogen-ion concentration, even though its pH is only 2 units higher.

Sound intensity level. The decibel level of sound intensity I relative to a reference intensity I_0 is

$$L = 10 \log_{10} \left(\frac{I}{I_0} \right).$$

Thus a 10-decibel increase multiplies intensity by 10, while a 20-decibel increase multiplies intensity by 100. Perceived loudness is more complicated, but the intensity calculation follows this logarithmic rule.

Earthquake magnitude. The Richter magnitude of an earthquake with seismic wave amplitude A , relative to a reference amplitude A_0 , is

$$M = \log_{10} \left(\frac{A}{A_0} \right).$$

Each whole-number increase in magnitude corresponds to a 10-fold increase in wave amplitude: a magnitude-7 earthquake has 10 times the amplitude of a magnitude-6 earthquake, and 100 times the amplitude of a magnitude-5 earthquake.

Solving for time in a growth model. A savings account with continuous compounding follows $A(t) = Pe^{rt}$. If the target balance A is known, logarithms isolate the unknown time:

$$A = Pe^{rt} \implies \frac{A}{P} = e^{rt} \implies \ln \left(\frac{A}{P} \right) = rt \implies \boxed{t = \frac{\ln(A/P)}{r}}.$$

Example (calculator). How long does it take \$2000 invested at 6% annual interest, compounded continuously, to grow to \$3000?

$$t = \frac{\ln(3000/2000)}{0.06} = \frac{\ln(1.5)}{0.06} \approx 6.758.$$

It takes about 6.76 years.

Exercise 14 (pH). A solution has $[\text{H}^+] = 10^{-6.2}$. Find its pH. How many times greater is its hydrogen-ion concentration than a solution with pH 8.2?

Exercise 15 (decibels). A sound changes from 50 dB to 80 dB. By what factor does its intensity change? Explain using the formula for L .

Exercise 16 (calculator application). A \$5000 investment earns 5% annual interest compounded continuously. Use logarithms to find how long it takes to reach \$8000. Round to two decimal places.

Exercise 17 (modeling, calculator). A bacteria culture follows $P(t) = 300 \cdot 2^{t/4}$, where t is measured in hours. Use logarithms to find when the population first reaches 5000. Give an exact logarithmic expression and a calculator approximation in hours.

Exercise 18 (challenge). Let $f(x) = a^x$ for $a > 0$, $a \neq 1$. Prove that $f^{-1}(x) = \log_a x$ directly by checking both compositions, including the necessary domain restrictions.